

Nicholas Saunders

Postdoctoral Fellow, Yale University

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Education

2021–2025	PhD	Astronomy , University of Hawai‘i at Mānoa Dissertation: “ <i>Stellar Evolution & Giant Exoplanets</i> ” Advisors: Daniel Huber & Jennifer van Saders
2019–2021	MSc	Astronomy , University of Hawai‘i at Mānoa
2013–2018	BS	Physics & Astronomy (<i>with Honors</i>), University of Washington
2013–2018	BA	Comparative Literature (<i>Cinema Studies</i>), University of Washington

Academic Positions

2025–	YCAA Prize Postdoctoral Fellow , Yale University
2021–2025	NSF Graduate Research Fellow , University of Hawai‘i at Mānoa
2021–2024	Visiting Scientist , American Museum of Natural History
2016–2018	Undergraduate Research Assistant , University of Washington
2011–2013	High School Student Astronomy Researcher , The Bear Fight Institute

Relevant Employment

2020–2020	Kepler & K2 Training Materials Developer , NumFocus, The Astropy Project
2018–2019	Kepler & K2 Research Support Scientist , NASA Ames Research Center
2018–2018	Data Visualization Analyst , University of Washington Mobile Planetarium

Publication Summary

[NASA ADS Library](#) | A full publication list can also be found below.

40 total publications (**5** first author, **8** significant contributions)

7,400+ total citations | h-index: **17**

Honors

2025	Yale Center for Astronomy & Astrophysics Prize Postdoctoral Fellowship
2025	Student Excellence in Research Award, University of Hawai‘i
2024	Achievement Rewards for College Scientists (ARCS)
2024	ARCS Scholar of the Year
2023	Travel Award, Aspen Center for Physics Winter Workshop
2021	NSF Graduate Research Fellowship
2019	Honorable Mention, NSF Graduate Research Fellowship
2018	Departmental Honors in Astronomy, University of Washington
2016	Best Online Photo Essay, Washington Newspaper Publishers Association
2013	Washington NASA Space Grant

Grants & Proposals

Telescope Time — 19.5 total nights awarded as PI, 90+ total nights spent on-sky

2026	PI, 4.0n Palomar/WIRC	Yale TAC: 26B_Y02
2023–2026	PI, 8.0n Keck-I/KPF	Yale TAC: 26A_Y233, 26B_Y205 UH TAC: 23A_H250, 23B_H390, 24A_H391, 24B_H344
2022–2025	PI, 7.5n Keck-I/HIRES	UH TAC: 22B_H266, 22B_H267, 23A_H247, 23A_H249, 23B_H389, 24A_H393, 24B_H343, 25A_H387
2026	Co-I, 0.5n Keck-I/KPF	Yale TAC: 26A_Y227
2021–2026	Co-I, 6.0n Keck-I/HIRES	Yale TAC: 26A_Y024 NASA TAC: 21A_N098, 21B_N085, 22A_N093, 22B_N025, 23A_N085

Funding

2020–2024	Co-I, \$190,000 TESS GI	PI S. Grunblatt: G03151, G04179, G07182
2021–2023	Co-I, \$70,512 NASA-Keck	PI S. Grunblatt: 21A_N098, 21B_N085, 22A_N093, 22B_N025, 23A_N085

Software Tools

Lightcurve: A friendly package for space telescope time series
Core Developer | **Tutorial Author** ([link](#)) | 503 GitHub ★s

Giants: TESS FFI pipeline for evolved exoplanet host stars
Solo Developer | [Paper](#)

Eleanor: Simple light curve extraction from TESS FFIs
Core Developer | [Paper](#) | 101 GitHub ★s

Scope: Simulated space telescope observations
Solo Developer | [Paper](#)

Presentations

Invited Talks

2026	Wesleyan University Astronomy Colloquium
2024	Princeton University Exoplanet Seminar
2024	MIT Special Seminar
2024	Yale University Stars & Planets Seminar
2024	CU Boulder APS Seminar (<i>remote</i>)
2023	Johns Hopkins University, STScI ESPF Seminar
2019	UC Berkeley CIPS Seminar
2017	NASA Ames Kepler/K2 Guest Observer Office Special Seminar

Contributed Talks

2026	New York Area Exoplanets Meeting, AMNH
2024	TESS Science Conference III, MIT
2024	Connecticut Exoplanet Picnic, Wesleyan University
2023	TASC7 / KASC14, University of Hawai'i

2023	AAS 241, Seattle, WA ×2
2021	TESS Science Team Meeting #25 (<i>remote</i>)
2020	AAS 235, Honolulu, HI
2019	AAS 233, Seattle, WA
2018	Kepler/K2 Science Office, NASA Ames
2018	Undergraduate Research Symposium, University of Washington
2018	AAS 231, Washington DC
2017	Undergraduate Research Symposium, University of Washington

Posters

2023	TASC7 / KASC14, Honolulu, HI
2023	Late-Stage and Post-Main-Sequence Systems Workshop, Aspen, CO ×2
2022	TASC6 / KASC13, Leuven, Belgium
2022	Cool Stars 21, Toulouse, France
2022	Exoplanets IV, Las Vegas, NV
2019	Kepler SciCon V, Glendale, CA
2019	AAS 233, Seattle, WA

Teaching & Outreach

Public Outreach

2023	HI STAR Summer Program, UH Maui College, Kahului, HI
2022	Hawai'i Public Radio Interview
2020–2021	Science Pen Pal, Letters to a Pre-Scientist
2019	Astronomy on Tap, San Jose, CA
2018	Astronomy on Tap, Seattle, WA
2018	Planetarium Organizer, University of Washington
2018	Astrobiology Mini Talks, Museum of Flight, Seattle, WA
2018	Astrobiology Mobile Planetarium Presenter, University of Washington
2015–2018	Weekly Public Planetarium Shows, University of Washington
2017	Volunteer, NASA Total Solar Eclipse Outreach Event

Guest Lectures

2026	ASTR 16, Williams College (<i>remote</i>)
2025	ASTR 575, Yale University
2019	ASTR 110, University of Hawai'i

Teaching

2019–2020	Graduate Teaching Assistant, ASTR 110, University of Hawai'i
2017–2018	Undergraduate Teaching Assistant, ASTR 160, University of Washington

Academic Advising

Undergraduate

2023	Advisor , Alicia Chun, Research Experience for Undergraduates, University of Chicago
2019–2021	Co-advisor , Erica Sawczynec, Undergraduate Honors Thesis, UH Mānoa

High School Students & Teachers

2025–	Advisor , Vihaan Iyer, NJ Ridge High School Student
2023	Advisor , Anica Ancheta, Dominic Rice, Holden Suzuki, HI STAR Research, HI High School Students
2022	Co-advisor , Alison English, Research Experience for Teachers, Honoka‘a High School Teacher
2020	Advisor , Wilson Chau, Pono Fortune, Gabe Mckillip, HI STAR Research, HI High School Students

Academic Service

Journal Reviewer

2026	Astronomy & Astrophysics
2025–2026	AAS Journals ×3
2025	Monthly Notices of the Royal Astronomical Society

Panels & Telescope Time Allocation Committees

2025–2026	Reviewer, Yale Astronomy TAC ×2
	Reviewer, NOIRLab Gemini Large Long Program TAC
	Reviewer, NSF Astronomy & Astrophysics Research Grant (AAG)
	Reviewer, NASA Exoplanet Research Program (XRP)

Local Organizing Committees

2026	<i>A Dynamical Synthesis of Planetary Systems</i> , Yale University
2023	<i>TESS/Kepler Asteroseismic Science Consortium</i> , University of Hawai‘i

Department Service

2024–2025	Cohort Kumukahi, Native Hawaiian Place of Learning, University of Hawai‘i
2022–2023	Graduate Student Representative, University of Hawai‘i
2020–2021	Graduate Outreach Representative, University of Hawai‘i

Publications

First-Author Publications

1. **N. Saunders**, S. K. Grunblatt, D. Huber, J. M. J. Ong, et al. “TESS Giants Transiting Giants. VII. A Hot Saturn Orbiting an Oscillating Red Giant Star”. In: *AJ* 169.2, 75 (Feb. 2025), p. 75. doi: [10.3847/1538-3881/ad9a87](https://doi.org/10.3847/1538-3881/ad9a87).
2. **N. Saunders**, S. K. Grunblatt, A. Chontos, F. Dai, et al. “TESS Giants Transiting Giants. VI. Newly Discovered Hot Jupiters Provide Evidence for Efficient Obliquity Damping after the Main Sequence”. In: *AJ* 168.2, 81 (Aug. 2024), p. 81. doi: [10.3847/1538-3881/ad543b](https://doi.org/10.3847/1538-3881/ad543b).
3. **N. Saunders**, J. L. van Saders, A. J. Lyttle, T. S. Metcalfe, et al. “Stellar Cruise Control: Weakened Magnetic Braking Leads to Sustained Rapid Rotation of Old Stars”. In: *ApJ* 962.2, 138 (Feb. 2024), p. 138. doi: [10.3847/1538-4357/ad1516](https://doi.org/10.3847/1538-4357/ad1516).
4. **N. Saunders**, S. K. Grunblatt, D. Huber, K. A. Collins, et al. “TESS Giants Transiting Giants. I.: A Noninflated Hot Jupiter Orbiting a Massive Subgiant”. In: *AJ* 163.2, 53 (Feb. 2022), p. 53. doi: [10.3847/1538-3881/ac38a1](https://doi.org/10.3847/1538-3881/ac38a1).

5. **N. Saunders**, R. Luger, and R. Barnes. “The Pointing Limits of Transiting Exoplanet Light Curve Characterization with Pixel Level Decorrelation”. In: *AJ* 157.5, 197 (May 2019), p. 197. doi: [10.3847/1538-3881/ab12e4](https://doi.org/10.3847/1538-3881/ab12e4).

Significant Contributions

6. S. K. Grunblatt, **N. Saunders**, D. Huber, D. Thorngren, et al. “TESS Giants Transiting Giants. IV. A Low-density Hot Neptune Orbiting a Red Giant Star”. In: *AJ* 168.1, 1 (July 2024), p. 1. doi: [10.3847/1538-3881/ad4149](https://doi.org/10.3847/1538-3881/ad4149).
7. S. K. Grunblatt, **N. Saunders**, A. Chontos, S. Hattori, et al. “TESS Giants Transiting Giants. III. An Eccentric Warm Jupiter Supports a Period-Eccentricity Relation for Giant Planets Transiting Evolved Stars”. In: *AJ* 165.2, 44 (Feb. 2023), p. 44. doi: [10.3847/1538-3881/aca670](https://doi.org/10.3847/1538-3881/aca670).
8. D. Stello, **N. Saunders**, S. Grunblatt, M. Hon, et al. “TESS asteroseismology of the Kepler red giants”. In: *MNRAS* 512.2 (May 2022), pp. 1677–1686. doi: [10.1093/mnras/stac414](https://doi.org/10.1093/mnras/stac414).
9. S. K. Grunblatt, **N. Saunders**, M. Sun, A. Chontos, et al. “TESS Giants Transiting Giants. II. The Hottest Jupiters Orbiting Evolved Stars”. In: *AJ* 163.3, 120 (Mar. 2022), p. 120. doi: [10.3847/1538-3881/ac4972](https://doi.org/10.3847/1538-3881/ac4972).
10. C. Hedges, **N. Saunders**, and J. Martínez-Palomera. “Contaminante: A Tool for Automatically Finding a Close-to-optimal Aperture for Transiting Signals in Kepler, K2, and TESS Data”. In: *Research Notes of the American Astronomical Society* 5.11, 260 (Nov. 2021), p. 260. doi: [10.3847/2515-5172/ac3765](https://doi.org/10.3847/2515-5172/ac3765).
11. C. Hedges, **N. Saunders**, G. Barentsen, J. L. Coughlin, et al. “Four Small Planets Buried in K2 Systems: What Can We Learn for TESS?” In: *ApJL* 880.1, L5 (July 2019), p. L5. doi: [10.3847/2041-8213/ab2a74](https://doi.org/10.3847/2041-8213/ab2a74).
12. C. Hedges, **N. Saunders**, J. L. Coughlin, G. Barentsen, et al. “A Transiting Hot Jupiter Candidate toward the Galactic Center Identified in the Kepler/K2 Campaign 9 Microlensing Survey”. In: *Research Notes of the American Astronomical Society* 3.1, 18 (Jan. 2019), p. 18. doi: [10.3847/2515-5172/aaff6f](https://doi.org/10.3847/2515-5172/aaff6f).
13. G. Barentsen, C. Hedges, **N. Saunders**, A. M. Cody, et al. “Kepler’s Discoveries Will Continue: 21 Important Scientific Opportunities with Kepler & K2 Archive Data”. In: *arXiv e-prints*, arXiv:1810.12554 (Oct. 2018), arXiv:1810.12554. doi: [10.48550/arXiv.1810.12554](https://doi.org/10.48550/arXiv.1810.12554).

Co-Author Publications

14. Y. Li, M. C. Liu, T. J. Dupuy, D. Huber, et al. “A Test of Substellar Evolutionary Models with High-precision Ages from Asteroseismology and Gyrochronology for the Benchmark System HR 7672AB”. In: *ApJ* 1002.1, 13 (May 2026), p. 13. doi: [10.3847/1538-4357/ae4bdb](https://doi.org/10.3847/1538-4357/ae4bdb).
15. Y. Zhang, J. P. Wardenier, A. Householder, T. D. Komacek, et al. “Extreme Winds on the Emerging Dayside of an Ultrahot Jupiter”. In: *ApJL* 997.2, L40 (Feb. 2026), p. L40. doi: [10.3847/2041-8213/ae36a1](https://doi.org/10.3847/2041-8213/ae36a1).
16. J. Gomez Barrientos, H. A. Knutson, M. Sidel, M. Greklek-McKeon, et al. “From Earths to Super-Earths: Five New Small Planets Transiting M Dwarf Stars”. In: *AJ* 171.2, 99 (Feb. 2026), p. 99. doi: [10.3847/1538-3881/ae246f](https://doi.org/10.3847/1538-3881/ae246f).

17. L. Sha, A. M. Vanderburg, C. X. Huang, S. Christian, et al. “The Occurrence Rate of Nearby Planetary Companions to Hot Jupiters”. In: *arXiv e-prints*, arXiv:2601.13302 (Jan. 2026), arXiv:2601.13302. doi: [10.48550/arXiv.2601.13302](https://doi.org/10.48550/arXiv.2601.13302).
18. Y. Li, M. C. Liu, T. J. Dupuy, D. Huber, et al. “A Test of Substellar Evolutionary Models with High-Precision Ages from Asteroseismology and Gyrochronology for the Benchmark System HR 7672AB”. In: *arXiv e-prints*, arXiv:2512.06083 (Dec. 2025), arXiv:2512.06083. doi: [10.48550/arXiv.2512.06083](https://doi.org/10.48550/arXiv.2512.06083).
19. C. L. Brinkman, L. M. Weiss, D. Huber, R. A. Lee, et al. “The Compositions of Rocky Planets in Close-in Orbits Tend to Be Earth-like”. In: *AJ* 170.2, 109 (Aug. 2025), p. 109. doi: [10.3847/1538-3881/ade677](https://doi.org/10.3847/1538-3881/ade677).
20. Y. Li, D. Huber, J. M. J. Ong, J. van Saders, et al. “K Dwarf Radius Inflation and a 10 Gyr Spin-down Clock Unveiled through Asteroseismology of HD 219134 from the Keck Planet Finder”. In: *ApJ* 984.2, 125 (May 2025), p. 125. doi: [10.3847/1538-4357/adc737](https://doi.org/10.3847/1538-4357/adc737).
21. R. A. Lee, F. Dai, A. W. Howard, S. Halverson, et al. “TOI-6324 b: An Earth-mass Ultra-short-period Planet Transiting a Nearby M Dwarf”. In: *ApJL* 983.2, L36 (Apr. 2025), p. L36. doi: [10.3847/2041-8213/adadd7](https://doi.org/10.3847/2041-8213/adadd7).
22. J. Zhang, L. M. Weiss, D. Huber, J. W. Xuan, et al. “Discovery of a Jupiter Analog Misaligned to the Inner Planetary System in HD 73344”. In: *AJ* 169.4, 200 (Apr. 2025), p. 200. doi: [10.3847/1538-3881/ada60a](https://doi.org/10.3847/1538-3881/ada60a).
23. J. Zhang, D. Huber, L. M. Weiss, J. W. Xuan, et al. “A Testbed for Tidal Migration: The 3D Architecture of an Eccentric Hot Jupiter HD 118203 b Accompanied by a Possibly Aligned Outer Giant Planet”. In: *AJ* 168.6, 295 (Dec. 2024), p. 295. doi: [10.3847/1538-3881/ad86c4](https://doi.org/10.3847/1538-3881/ad86c4).
24. T. S. Metcalfe, J. L. van Saders, D. Huber, D. Buzasi, et al. “TESS Asteroseismology of β Hydri: A Subgiant with a Born-again Dynamo”. In: *ApJ* 974.1, 31 (Oct. 2024), p. 31. doi: [10.3847/1538-4357/ad6dd6](https://doi.org/10.3847/1538-4357/ad6dd6).
25. H. Isaacson, A. W. Howard, B. Fulton, E. A. Petigura, et al. “The California Legacy Survey. V. Chromospheric Activity Cycles in Main-sequence Stars”. In: *ApJS* 274.2, 35 (Oct. 2024), p. 35. doi: [10.3847/1538-4365/ad676c](https://doi.org/10.3847/1538-4365/ad676c).
26. F. Dai, A. W. Howard, S. Halverson, J. Orell-Miquel, et al. “An Earth-sized Planet on the Verge of Tidal Disruption”. In: *AJ* 168.3, 101 (Sept. 2024), p. 101. doi: [10.3847/1538-3881/ad5a7d](https://doi.org/10.3847/1538-3881/ad5a7d).
27. S. W. Yee, E. A. Petigura, H. Isaacson, A. W. Howard, et al. “Additional Doppler Monitoring Corroborates HAT-P-11c as a Planet”. In: *Research Notes of the American Astronomical Society* 8.7, 187 (July 2024), p. 187. doi: [10.3847/2515-5172/ad675e](https://doi.org/10.3847/2515-5172/ad675e).
28. N. L. Eisner, S. K. Grunblatt, O. Barragán, T. H. Faridani, et al. “Planet Hunters TESS. V. A Planetary System Around a Binary Star, Including a Mini-Neptune in the Habitable Zone”. In: *AJ* 167.5, 241 (May 2024), p. 241. doi: [10.3847/1538-3881/ad1d5c](https://doi.org/10.3847/1538-3881/ad1d5c).
29. A. Chontos, D. Huber, S. K. Grunblatt, **N. Saunders**, et al. “The TESS-Keck Survey XXI: 13 New Planets and Homogeneous Properties for 21 Subgiant Systems”. In: *arXiv e-prints*, arXiv:2402.07893 (Feb. 2024), arXiv:2402.07893. doi: [10.48550/arXiv.2402.07893](https://doi.org/10.48550/arXiv.2402.07893).

30. F. Pereira, S. K. Grunblatt, A. Psaridi, T. L. Campante, et al. “TESS giants transiting giants V - two hot Jupiters orbiting red giant hosts”. In: *MNRAS* 527.3 (Jan. 2024), pp. 6332–6345. doi: [10.1093/mnras/stad3449](https://doi.org/10.1093/mnras/stad3449).
31. D. R. Hey, D. Huber, B. J. Shappee, J. Bland-Hawthorn, et al. “The Far Side of the Galactic Bar/Bulge Revealed through Semi-regular Variables”. In: *AJ* 166.6, 249 (Dec. 2023), p. 249. doi: [10.3847/1538-3881/ad01bf](https://doi.org/10.3847/1538-3881/ad01bf).
32. S. Vissapragada, A. Chontos, M. Greklek-McKeon, H. A. Knutson, et al. “The Possible Tidal Demise of Kepler’s First Planetary System”. In: *ApJL* 941.2, L31 (Dec. 2022), p. L31. doi: [10.3847/2041-8213/aca47e](https://doi.org/10.3847/2041-8213/aca47e).
33. Astropy Collaboration, A. M. Price-Whelan, P. L. Lim, N. Earl, et al. “The Astropy Project: Sustaining and Growing a Community-oriented Open-source Project and the Latest Major Release (v5.0) of the Core Package”. In: *ApJ* 935.2, 167 (Aug. 2022), p. 167. doi: [10.3847/1538-4357/ac7c74](https://doi.org/10.3847/1538-4357/ac7c74).
34. S. K. Grunblatt, J. C. Zinn, A. M. Price-Whelan, R. Angus, et al. “Age-dating Red Giant Stars Associated with Galactic Disk and Halo Substructures”. In: *ApJ* 916.2, 88 (Aug. 2021), p. 88. doi: [10.3847/1538-4357/ac0532](https://doi.org/10.3847/1538-4357/ac0532).
35. C. Hedges, R. Angus, G. Barentsen, **N. Saunders**, et al. “Systematics-insensitive Periodogram for Finding Periods in TESS Observations of Long-period Rotators”. In: *Research Notes of the American Astronomical Society* 4.12, 220 (Dec. 2020), p. 220. doi: [10.3847/2515-5172/abd106](https://doi.org/10.3847/2515-5172/abd106).
36. A. D. Feinstein, B. T. Montet, D. Foreman-Mackey, M. E. Bedell, et al. “eleanor: An Open-source Tool for Extracting Light Curves from the TESS Full-frame Images”. In: *PASP* 131.1003 (Sept. 2019), p. 094502. doi: [10.1088/1538-3873/ab291c](https://doi.org/10.1088/1538-3873/ab291c).
37. T. J. David, A. M. Cody, C. L. Hedges, E. E. Mamajek, et al. “A Warm Jupiter-sized Planet Transiting the Pre-main-sequence Star V1298 Tau”. In: *AJ* 158.2, 79 (Aug. 2019), p. 79. doi: [10.3847/1538-3881/ab290f](https://doi.org/10.3847/1538-3881/ab290f).
38. A. Mahabal, U. Rebbapragada, R. Walters, F. J. Masci, et al. “Machine Learning for the Zwicky Transient Facility”. In: *PASP* 131.997 (Mar. 2019), p. 038002. doi: [10.1088/1538-3873/aaf3fa](https://doi.org/10.1088/1538-3873/aaf3fa).
39. A. M. Cody, G. Barentsen, C. Hedges, M. Gully-Santiago, et al. “A Catalog of 29 Open Clusters and Associations Observed by the Kepler and K2 Missions”. In: *Research Notes of the American Astronomical Society* 2.4, 199 (Nov. 2018), p. 199. doi: [10.3847/2515-5172/aaec76](https://doi.org/10.3847/2515-5172/aaec76).
40. R. Luger, E. Kruse, D. Foreman-Mackey, E. Agol, and **N. Saunders**. “An Update to the EVEREST K2 Pipeline: Short Cadence, Saturated Stars, and Kepler-like Photometry Down to $K_p = 15$ ”. In: *AJ* 156.3, 99 (Sept. 2018), p. 99. doi: [10.3847/1538-3881/aad230](https://doi.org/10.3847/1538-3881/aad230).

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